

Exercise on Sets

1. Create an empty set.
2. Create a set for numbers 1, 2, 3, 4, 5, 2, 1, 4, 3, 2, 3, 1, 5, 5, 2, 3, 1.
 - (i) using set().
 - (ii) If they were in list
 - (iii) If they were in tuple
3. Create a set for fruits ['apple', 'banana', 'berry', 'pineapple', 'banana', 'apple'].
4. Construct a set using each fruit listed in 3.
5. Construct a set for 'aaryan' and 'Aaryan'. Note down the difference.
6. What is the difference between set and frozenset.
7. Create a set for a tuple (1, 2, 3, 4, 5).
 - (i) Change the integer 3 to 33. Analyse the output given.
 - (ii) Include the integer '33' to the set created
 - (iii) Include the two integers '444' and '555' to the set in one-line command.
8. What is the difference between add() and update() in sets.
9. Can you access the third element of a set having 5 members?. Justify your answer.
10. A set has 4 fruits and a vegetable and you know the names of members. Remove the vegetable from the set.[construct a set and do this].
11. Construct a set as in Q10. Remove the vegetable using a method which is not used in Q10.

12. What is the difference of the methods used in Q10 and Q11?.
13. Construct a set of your choice having 5 elements and use a method to remove any one item from the set.
14. If the method used in Q13 is randomized, List at least one real life situation when it can be used.
15. Construct a set containing 6 elements and you remove all the items.
16. Consider, you went to a vegetable market and you purchased vegetables as many as you wished [you may have bought the same item from different shops in that case you have to include it separately]. You make a list of those. Print the total number of distinct items.
17. Your mother ask whether you have purchased a particular vegetable when you completed the Q16. Don't look at the list you prepared. Make a code to find it.
18. You and your friend have been assigned a task to prepare a list of exam centers across your state but you have not contacted him till you prepare the list. He / She came up with a list(soft copy)[Huge number of centers]. Check if your list and your friend's list share any common centers. [For simplicity you prepare a small list for understanding].
19. You are assigned to prepare a syllabus of a course and you have prepared the topic wise syllabus. Suddenly, you are asked that the syllabus should not deviate from a master list. Check if your syllabus is fully contained within the master syllabus. Construct a master syllabus and your syllabus and demonstrate(few topics).
20. Consider that you are a teacher coordinating an event and 4 different competition is going to be conducted. Prepare a student profile for each competition having 5 students for each event with name as keys and class as values. Prepare the dictionary in such a way that few students are participating

in more than one event. How will you find total number of lunch to be arranged for the participants?. [Result should be a program output].

21. You are shortlisting candidates for 2 different posts. It was instructed that a candidate can apply only for one post otherwise candidature will be rejected. You have received 15 application for each post. Create a list of name of the candidates for each post and consider 3 have applied for both posts. You find 3 names to be rejected.
22. A selection list prepared for a job. Create a list of 10 names of your choice, who are selected for the job. During police verification, the police has submitted a list of 6 names(not all who were selected) against whom at least one case is pending. Consider to include 2 names in the police list who are in the selection list. How will you find those two names to exclude from the list.
23. A college is going to conduct counselling for admission to physics and computer science course. Prepare name list of candidates selected for both courses 10 each. Prepare the list in such a way that 5 names appear in both the lists. Due to some reasons, the college withheld the students who were selected for both courses and decided to call only others for counselling. Prepare the list of candidates to be called for counselling.